

7.2 Natural Logarithms

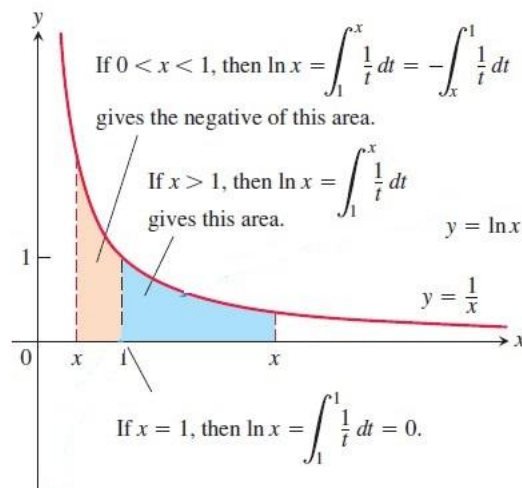
Historically, logarithms played important roles in arithmetic computations, making possible the great seventeenth-century advances in offshore navigation and celestial mechanics. In this section we define the natural logarithm as an integral using the Fundamental Theorem of Calculus. While this indirect approach may at first seem strange, it provides an elegant and rigorous way to obtain the key characteristics of logarithmic and exponential functions.

① Definition of the Natural Logarithm Function

DEFINITION The **natural logarithm** is the function given by

$$\ln x = \int_1^x \frac{1}{t} dt, \quad x > 0.$$

◆ From the Fundamental Theorem of Calculus, $\ln x$ is a continuous function.



◆ Geometrically, we can see the following results.

- a) If $x > 1$, then $\ln x$ is the area under the curve $y = \frac{1}{t}$ from $t = 1$ to $t = x$. So, $\ln x > 0$ if $x > 1$.

b) For $0 < x < 1$, $\ln x$ gives the negative of the area under the curve from x to 1.

So, $\ln x < 0$ if $0 < x < 1$ and the function is not defined for $x \leq 0$.

◆ $\ln 1 = \int_1^1 \frac{1}{t} dt = 0.$

◆ By using rectangles to obtain finite approximations of the area under the graph of $y = \frac{1}{t}$ and over the interval between $t = 1$ and $t = x$, as in Section 5.1, we can approximate the values of the function $\ln x$. Several values are given in the following Table.

x	$\ln x$
0	undefined
0.05	-3.00
0.5	-0.69
1	0
2	0.69
3	1.10
4	1.39
10	2.30

There is an important number between $x = 2$ and $x = 3$ whose natural logarithm equals 1. This number, which we now define, exists because $\ln x$ is a continuous function and therefore satisfies the Intermediate Value Theorem on $[2, 3]$.

DEFINITION The **number e** is the number in the domain of the natural logarithm that satisfies

$$\ln e = \int_1^e \frac{1}{t} dt = 1.$$

In the next section, we will see that the number e can be calculated as a limit and has the numerical value $e \approx 2.718281828459045$ to 15 decimal places.

② The Derivative of $y = \ln x$

◆ By the first part of the Fundamental Theorem of Calculus, we get

$$(\ln x)' = \left(\int_1^x \frac{1}{t} dt \right)' = \frac{1}{x}, \quad x > 0.$$

◆ The Chain Rule about natural logarithm is

$$(\ln u)' = \frac{1}{u} \frac{du}{dx},$$

where $u = u(x)$.

Note Notice that the derivative is always positive. This means that $y = \ln x$ is always increasing.

Example 1 Find the derivatives.

- (a) $\ln 2x$;
- (b) $\ln (x^2 + 3)$;
- (c) $\ln |x|$.

③ Properties of Logarithms

THEOREM 2 — Algebraic Properties of the Natural Logarithm

For any numbers $b > 0$ and $x > 0$, the natural logarithm satisfies the following rules:

- 1. *Product Rule:* $\ln bx = \ln b + \ln x$
- 2. *Quotient Rule:* $\ln \frac{b}{x} = \ln b - \ln x$
- 3. *Reciprocal Rule:* $\ln \frac{1}{x} = -\ln x$
- 4. *Power Rule:* $\ln x^r = r \ln x$ (r rational)

Example 2

- (a) $\ln 6 = \ln 2 \cdot 3 = \ln 2 + \ln 3$.

$$(b) \ln 4 - \ln 5 = \ln \frac{4}{5} = \ln 0.8.$$

$$(c) \ln \frac{1}{8} = \ln 1 - \ln 8 = -\ln 8.$$

Example 3

$$(a) \ln 4 + \ln \sin x = \ln 4 \sin x.$$

$$(b) \ln \frac{x+1}{2x-3} = \ln(x+1) - \ln(2x-3).$$

$$(c) \ln \sqrt[3]{x+1} = \ln(x+1)^{\frac{1}{3}} = \frac{1}{3} \ln(x+1).$$

Example 4 Find y' if

$$(a) y = \frac{1}{2} \ln \frac{1+x}{1-x};$$

$$(b) y = \ln \sqrt{\frac{(1+x)^5}{(x+2)^7}}.$$

④ The Graph and Range of $\ln x$

The derivative $(\ln x)' = \frac{1}{x}$ is positive for $x > 0$, so $\ln x$ is an increasing function of x .

The second derivative, $(\ln x)'' = -\frac{1}{x^2}$, is negative, so the graph of $\ln x$ is concave down.

By ①, we know that $\ln 2 = 0.69 > \frac{1}{2}$. Hence, we have

$$\ln 2^n = n \ln 2 > n \left(\frac{1}{2} \right) = \frac{n}{2}.$$

This result shows that

$$\lim_{n \rightarrow \infty} \ln 2^n > \lim_{n \rightarrow \infty} \frac{n}{2} = \infty.$$

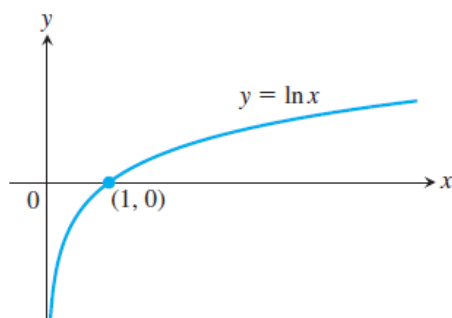
Since $\ln x$ is an increasing function, we get that

$$\lim_{x \rightarrow \infty} \ln x = \infty.$$

We also have

$$\lim_{x \rightarrow 0^+} \ln x = \lim_{t \rightarrow \infty} \ln \frac{1}{t} = \lim_{t \rightarrow \infty} (-\ln t) = -\infty.$$

We defined $\ln x$ for $x > 0$, so the domain of $\ln x$ is the set of positive real numbers. The above discussion and the Intermediate Value Theorem show that its range is the entire real line, giving the graph of $y = \ln x$ shown in the following Figure.



$$D = (0, \infty), \quad R = (-\infty, \infty).$$

⑤ The integral $\int \frac{1}{u} du$

If u is a differentiable function that is never zero, then

$$\int \frac{1}{u} du = \ln |u| + C.$$

Example 5 Evaluate

$$(a) \int \frac{2x}{x^2 - 5} dx; \quad (b) \int \frac{4 \cos x}{3 + 2 \sin x} dx.$$

Example 6 Evaluate

$$(a) \int \cot x dx; \quad (b) \int \tan x dx; \quad (c) \int \tan 2x dx;$$

⑥ Logarithmic Differentiation

The derivatives of positive functions given by formulas that involve products, quotients, and powers can often be found more quickly if we take the natural logarithm of both sides before differentiating. This enables us to use the laws of logarithms to simplify the formulas before differentiating. The process, called **logarithmic differentiation**, is

illustrated in the next example.

Example 7 Find $\frac{dy}{dx}$, if

$$y = \frac{(x^2 + 1)(x + 3)^{\frac{1}{2}}}{x - 1}.$$

Example 8 Find $\frac{dy}{dx}$, if

$$y = \frac{(x^3 + 2x)^3 (2x + 3)^{\frac{1}{2}}}{\sin x}.$$